Strings

int

character a A

b B

c c

z z

~

*

#

Sequence of characters

Str A = "Abhinav"

str B = "Atul"

char C = 'A'

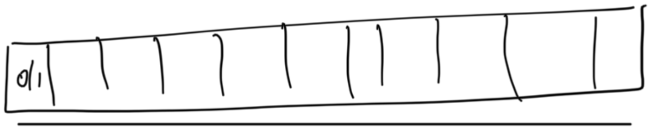
$\boxed{\text{char D = 'AB'}}$

":D"

4 byte ≡ int A

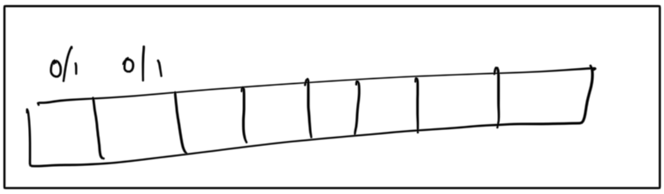
in

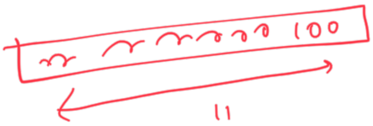
32 bits



char C

1 byte

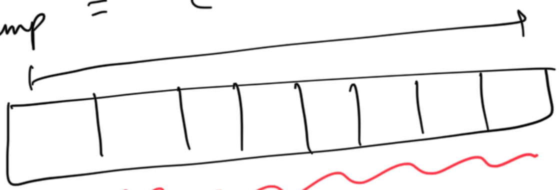




=

4

char temp = 'c'



1 byte = 8 bits

⇓

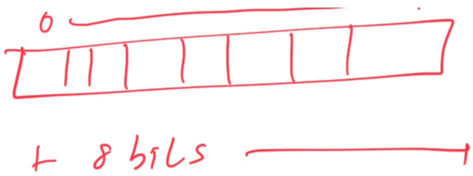
c

ASCII characters

0 - 255

0 - 255 → 65

'A' ←



ASCII

65 → 'A'
66 → 'B'
67 → 'C'

90 → 'Z'

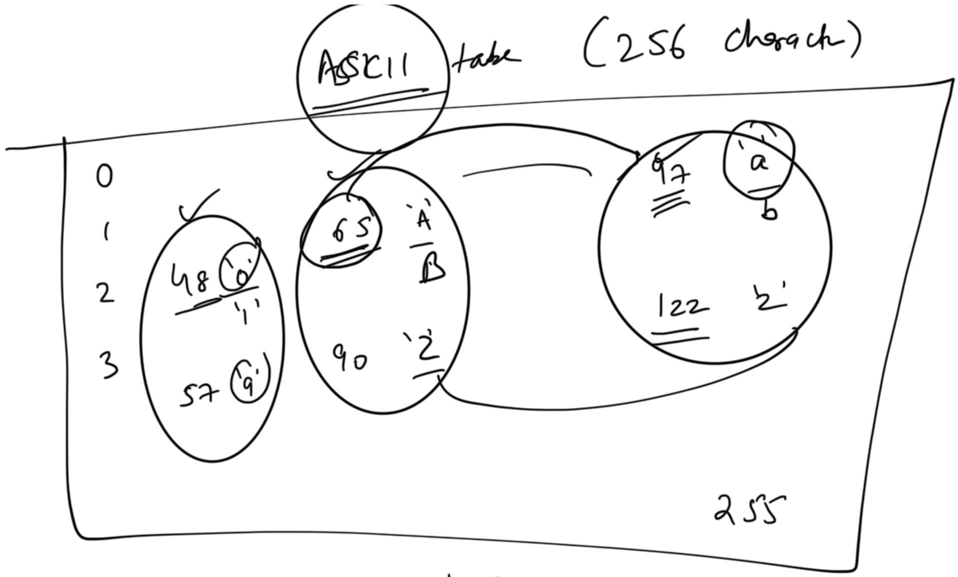
97 → 'a'
98 → 'b'

122 ← 'z'

48 – '0'
49 – '1'

57 – '9'

97 − 32 = 65
122 − 32 = 90



65 → 'A'
32 { 97 → 'a'

8 bits

'B'
66
98 'b'

0

00000000000000

'C'
67
99 'c'

'0'

a26 5 b c 6 8

string ≡ seq of chars

str A[10] → 10 bytes

H

Immutable vs Mutable

Java / Python

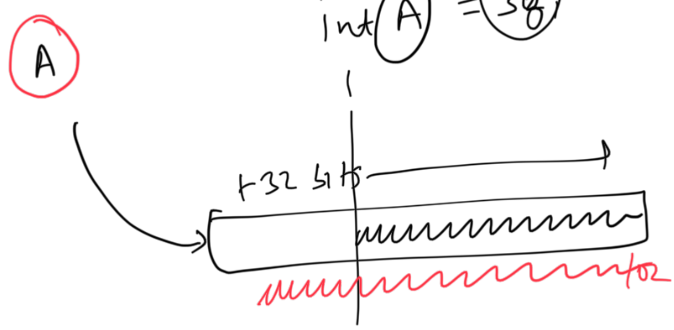
Immutable

str A = "Tarun"

A[0] = B

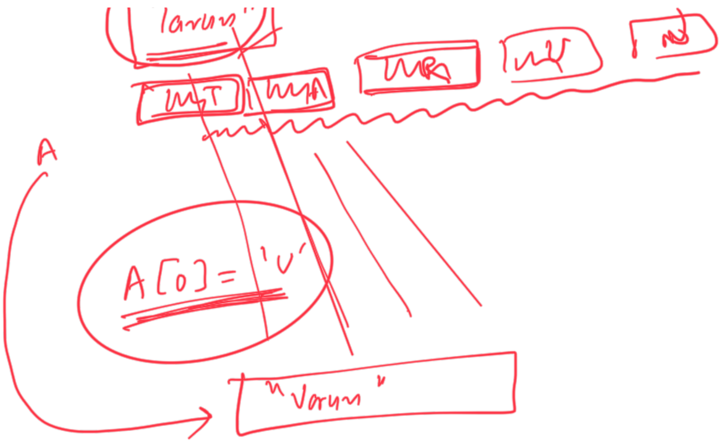
"Tarun" → "Barun"

int A = 38



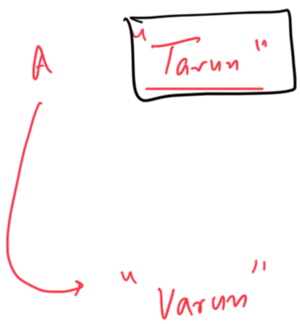
A = 102

str A = "Tarun"



$T.C = O(N)$

aux $S.C = O(N)$



garbage collection

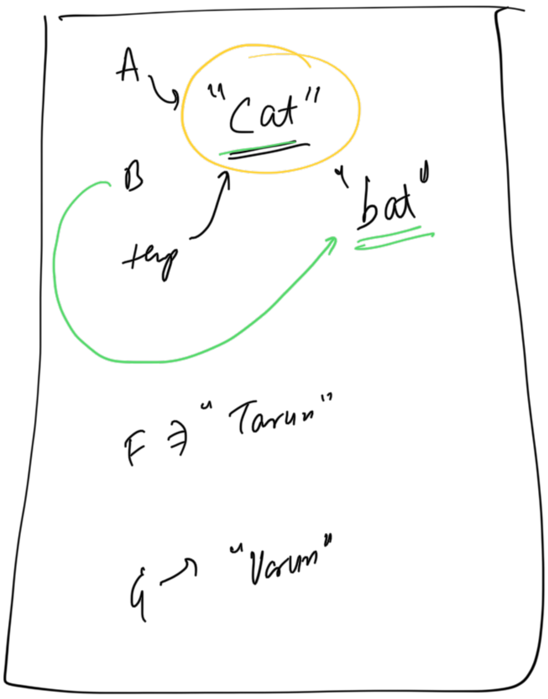
Implication

✓ (1) Modification of immutable string $\equiv O(N)$ time
↓
size of string

✓ (2) aux space $= O(N)$

✓ (3) garbage collection becomes intensive

(4) string pool memory which makes it efficient overall



String pool

Str A = "cat"
Str B = "cat"
Str temp = "cat"

B[0] = 'b' // O(N)

String s1 = "a"

s1 = s1 + 'b'

s1 = s1 + 'c'

T
k

O(kN)

If you get TLE for string manipulation, you should suspect upon immutable strings :)

I_am feeling lucky today :P

String Builder sb = new String Builder()

sb. add(a) < a >

sb.add(b) < a b >

vector

list

sb. add(c) < abc >

sb.to_string() → abc

Q S Toggle every character to the opposite case

S : abTFgKl

↓

s' : ABtfGkL

toggle(s)

{ sb = {}

O(N) for (i=0, i < S.size(), i++)

{ // lowerCase

if (s[i] >= 'a' && s[i] <= 'z')

{ ~~s[i] = s[i] - 32~~ sb.add(s[i]-32)

O(N)

}

else if (s[i] >= 'A' && s[i] <= 'Z')

{ ~~s[i] = s[i] + 32~~

sb.add(s[i]+32)

}

~~return s~~

return sb.toString();

}

O(N^2) time

O(N) time using StringBuilder

if immutable

O(N)

mutable string

Q2

String s of (lower case english characters)

97 97 => 0

100 − 97 => 3

s: dabaedb

(sort a)

☆ s' aabbdde

f(a) b c d e

2	2	0	2	1	0	0	0	0

O(N log N)

$\boxed{\text{aabbdde}}$

a 97

b

c

z 122

0 1 25

CountA | #(a) | #(b) | ... | #(z) |

1 - 26

COUNT SORT

If you know what all possibilities do you have for every character

0 1 2 3 25

| 1 | | | 1 | | |

int CountA [26]

// Create the count Array Loop!

O(N) time

```
for (i=0 ; i < s.size() ; i++)     O(N)
{
    int index = [s[i] - 97]   //s[i] - 'a'
    CountA[index]++;
}
```

'a' = 97 - 97 = 0

'b' 98 - 97 = 1

99 - 97 = 2

'z' 122 - 97 = 25

a b c z

97, 98, 99 ___ 122

-97 -97 -97 -97

0 1 2 25

sb = { }

i < 26 ; i++)

loop2 O(c) for(i=0; i<26; i++)

{

int count = count A[i] //a i=0

//b i=1

O(N) for (j=0; j<count; j++) //c i=2

sb.add((char)('a'+i))

}

return sb.toStr()

}

O(N) time

loop1 + loop2

O(N) O(N)

= T.C = O(N)

10:41 p

break

10:45 p

'a' + 0

char + int

ASCII int + int

(char) (ASCII int)

Q

S

s = | a | b | d | e | a | g | f | e |

l r

Reverse the substring b/w l and r

s' | a | b | f | g | a | e | d | e |

```
reverse(s, l, r)
{
    while( l < r)
    {
        swap( s(l), s(r))
        l++
```

(Q) Amazon

Sentence as a string

S = | h | e | r | e | _ | i | s | _ | a | _ | b | o | y |

* between every two words, we have a space character

☆ s' | b | o | y | _ | a | _ | i | s | _ | h | e | r | e |

words reversed in the sentence

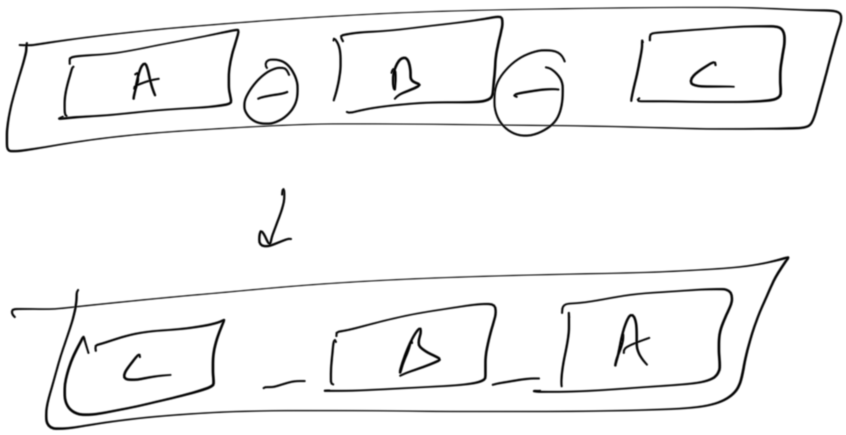
S : here_is_a_boy

s' : boy_a_is_here

S

s'

Obs1: size(input) = size(output)



S = " mailmen big letters"

0 1 2 3 4 5 6 7 8 9 10 11 12

S* sretteL gnizb nemlia

letters

big

main

S' = " letters big mailmen"

Obs(1): size(input) = size(output)

Obs(2): cognizant of the word boundary

S $\xrightarrow[\text{I}]{\text{reverse}}$ S*

II reverse every word

S Ans

$O(N)$ ① reverse$(0, N-1)$

$O(wordL)$ $O(wordL)$ $O(wordL)$

$T.C = O(N) + \cancel{O(WL1) + O(WL2) + O(WL3)}$
$\ldots$
$O(WLk)$

$O(N)$

$O(N) + O(N)$
$= O(N)$

aux $S.C = O(N)$

palindroms

Madam

mom

Nitin

aba

abba

$\boxed{abca}$

acba

~~Mala~~

Malayalam

N

$s = i\,f\boxed{abba}fgh\,i$

⓪ Find the longest palindromic substring in a given s

$s = mom$

$s' = mom$

3

$s = anitinb$

$s' = nitin$

5

S : acbalm

☆ 1

S: abacab

5 ☆